

Nakajima's representation of the Heisenberg algebra

Daniel González Casanova Azuela

PUC-Rio

November 28, 2025

Plan of the talk

Heisenberg Algebra

Hilbert Scheme of Points

Borel-Moore Homology

Correspondences

Main Construction

Idea of Proof

Heisenberg algebra

The Heisenberg algebra is $\hat{\mathfrak{a}} = \bigoplus_{n \in \mathbb{Z}} \mathbb{C} h_n \oplus \mathbb{C} K$ with bracket given by

$$[h_m, h_n] = m\delta_{m, -n}K, \quad [K, \hat{\mathfrak{a}}] = 0,$$

and highest weight representation

$$H = \mathbb{C}[x_1, x_2, \dots]$$

with the action

$$h_n \mapsto \begin{cases} n \frac{\partial}{\partial x_n} & \text{if } n > 0 \\ x_{-n} & \text{if } n < 0, \\ 0 & \text{if } n = 0 \end{cases}, \quad K \mapsto \text{Id.}$$

Hilbert scheme of points

Let X be a projective surface over $\mathbb{C}$. The essential property of the *Hilbert scheme of n points* $X^{[n]}$ is that a point (over $\mathbb{C}$) of $X^{[n]}$ is a 0-dimensional closed subscheme of X of length n .

Hilbert scheme of points

Let X be a projective surface over $\mathbb{C}$. The essential property of the *Hilbert scheme of n points* $X^{[n]}$ is that a point (over $\mathbb{C}$) of $X^{[n]}$ is a 0-dimensional closed subscheme of X of length n .

The *length* of any such subscheme Z , defined by $\dim \Gamma(\mathcal{O}_Z)$, is given by $\sum_{x \in \text{supp} Z} \dim \Gamma(\mathcal{O}_{Z_x}) = n$.

Hilbert scheme of points

Let X be a projective surface over $\mathbb{C}$. The essential property of the *Hilbert scheme of n points* $X^{[n]}$ is that a point (over $\mathbb{C}$) of $X^{[n]}$ is a 0-dimensional closed subscheme of X of length n .

The *length* of any such subscheme Z , defined by $\dim \Gamma(\mathcal{O}_Z)$, is given by $\sum_{x \in \text{supp} Z} \dim \Gamma(\mathcal{O}_{Z_x}) = n$.

This **does not** mean that the points in $X^{[n]}$ are n -tuples of points of X counted with multiplicities.

Hilbert scheme of points

Theorem (Fogarty)

If X is a nonsingular projective surface,

1. $X^{[n]}$ is nonsingular and has dimension $2n$, and
2. the Hilbert-Chow morphism

$$\begin{aligned}\pi : X^{[n]} &\longrightarrow S^n X \\ Z &\longmapsto \sum_{x \in \text{supp} Z} \dim \Gamma(\mathcal{O}_{Z,x})[x]\end{aligned}$$

is a resolution of singularities.

Borel-Moore homology

Borel-Moore homology $H_{\bullet}^{BM}(X)$ is the homology of the chain complex of **locally finite** linear combinations of singular simplices.

Borel-Moore homology

Borel-Moore homology $H_{\bullet}^{BM}(X)$ is the homology of the chain complex of **locally finite** linear combinations of singular simplices.

1. (Pushforward.) If $f : X \rightarrow Y$ is proper then there is a map $f_* : H_i^{BM}(X) \rightarrow H_i^{BM}(Y)$.

Borel-Moore homology

Borel-Moore homology $H_{\bullet}^{BM}(X)$ is the homology of the chain complex of **locally finite** linear combinations of singular simplices.

1. (Pushforward.) If $f : X \rightarrow Y$ is proper then there is a map $f_* : H_i^{BM}(X) \rightarrow H_i^{BM}(Y)$.
2. (Pullback.) If $f : X \rightarrow Y$ is a locally trivial fibration with fiber a manifold of dimension m (over $\mathbb{R}$), then there is a map $f^* : H_i^{BM}(Y) \rightarrow H_{i+m}^{BM}(X)$.

Borel-Moore homology

Borel-Moore homology $H_{\bullet}^{BM}(X)$ is the homology of the chain complex of **locally finite** linear combinations of singular simplices.

1. (Pushforward.) If $f : X \rightarrow Y$ is proper then there is a map $f_* : H_i^{BM}(X) \rightarrow H_i^{BM}(Y)$.
2. (Pullback.) If $f : X \rightarrow Y$ is a locally trivial fibration with fiber a manifold of dimension m (over $\mathbb{R}$), then there is a map $f^* : H_i^{BM}(Y) \rightarrow H_{i+m}^{BM}(X)$.
3. (Intersection pairing.) If X and Y are both “nicely embedded” in the same manifold M , there is a map

$$\cap_M : H_i^{BM}(X) \times H_j^{BM}(Y) \rightarrow H_{i+j-m}^{BM}(X \cap Y).$$

Borel-Moore homology

Borel-Moore homology $H_{\bullet}^{BM}(X)$ is the homology of the chain complex of **locally finite** linear combinations of singular simplices.

1. (Pushforward.) If $f : X \rightarrow Y$ is proper then there is a map $f_* : H_i^{BM}(X) \rightarrow H_i^{BM}(Y)$.
2. (Pullback.) If $f : X \rightarrow Y$ is a locally trivial fibration with fiber a manifold of dimension m (over $\mathbb{R}$), then there is a map $f^* : H_i^{BM}(Y) \rightarrow H_{i+m}^{BM}(X)$.
3. (Intersection pairing.) If X and Y are both “nicely embedded” in the same manifold M , there is a map

$$\cap_M : H_i^{BM}(X) \times H_j^{BM}(Y) \rightarrow H_{i+j-m}^{BM}(X \cap Y).$$

4. (Fundamental class.) If X is a variety of dimension d over $\mathbb{C}$ then there is *fundamental class* $[X] \in H_{2d}^{BM}(X)$.

Correspondences

Let X_1, X_2 be smooth irreducible varieties of dimensions d_1, d_2 . Let Z be a d -dimensional closed subvariety of $X_1 \times X_2$ such that the projections p_1 and p_2 are proper.

The *correspondence* associated to $Z \subset X_1 \times X_2$ is the map

$$\begin{aligned} c_{[Z]} : H_i^{BM}(X_2) &\longrightarrow H_{i+2d-2d_2}^{BM}(X_1) \\ c &\longmapsto p_{1,*}(p_2^*c \cap [Z]) \end{aligned}$$

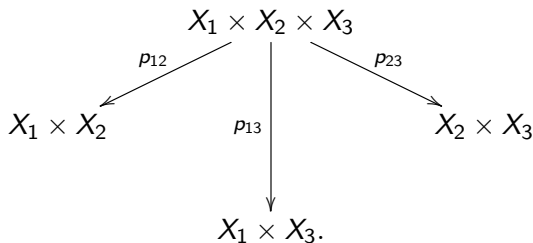
where p_1 and p_2 are the projections in the following diagram:

$$\begin{array}{ccc} & Z \subset X_1 \times X_2 & \\ p_1 \swarrow & & \searrow p_2 \\ X_1 & & X_2. \end{array}$$

Composition of correspondences

Let $Z_{12} \subset X_1 \times X_2$ and $Z_{23} \subset X_2 \times X_3$ such that the projections to X_2 and X_3 are proper. The composition of correspondences is

$$[Z_{12}] \circ [Z_{23}] := p_{13,*}(p_{12}^*[Z_{12}] \cap p_{23}^*[Z_{23}]),$$



In fact, the projection $p_{13}(p_{12}^{-1}Z_{12} \cap p_{23}^{-1}Z_{23}) \rightarrow X_3$ is proper.

Main construction

Let $i > 0$. Define $P[i] \subseteq \coprod_n X^{[n-i]} \times X^{[n]} \times X$ by

$$P[i] := \coprod_n \{(\mathcal{J}_1 \supseteq \mathcal{J}_2, x) : \text{supp}(\mathcal{J}_1/\mathcal{J}_2) = \{x\}\}.$$

Main construction

Let $i > 0$. Define $P[i] \subseteq \coprod_n X^{[n-i]} \times X^{[n]} \times X$ by

$$P[i] := \coprod_n \{(\mathcal{J}_1 \supseteq \mathcal{J}_2, x) : \text{supp}(\mathcal{J}_1/\mathcal{J}_2) = \{x\}\}.$$

Let $\alpha \in H_*^{BM}(X)$, $\beta \in H_*(X)$ and $i > 0$. Define

$$P_\alpha[i] = \varpi_*(\Pi_1^* \alpha \cap [P[i]])$$

$$P_\beta[i] = \varpi_*(\Pi_1^* \beta \cap [P[-i]]).$$

$$\begin{array}{ccc} & \coprod_n X^{[n-i]} \times X^{[n]} \times X & \\ \swarrow \Pi_1 & & \searrow \varpi_i \\ X & & \coprod_n X^{[n-i]} \times X^{[n]}. \end{array}$$

Main theorem

Theorem (Nakajima, Grojnowski)

If $(\deg \alpha)(\deg \beta) \equiv 0 \pmod{2}$,

$$[P_\alpha[i], P_\beta[j]] = (-1)^{i-1} i \delta_{i,-j} \langle \alpha, \beta \rangle \text{Id},$$

where $\langle \alpha, \beta \rangle = p_(\alpha \cap \beta)$ for $p : X \rightarrow \text{pt}$ the unique morphism from X to the point pt .*

Easy case

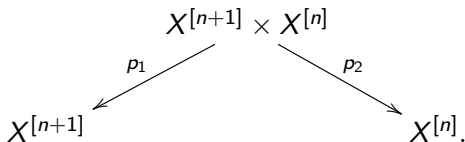
Let $x \in X$, $\alpha = [X]$, $\beta = [x]$. Take a set of distinct n points $x_1, \dots, x_n \in X \setminus \{x\}$ and identify them with the ideal $\mathcal{J}$. Then

$$[P_\alpha[1], P_\beta[-1]][\mathcal{J}] = [\mathcal{J}].$$

Easy case

Let $x \in X$, $\alpha = [X]$, $\beta = [x]$. Take a set of distinct n points $x_1, \dots, x_n \in X \setminus \{x\}$ and identify them with the ideal $\mathcal{J}$. Then

$$[P_\alpha[1], P_\beta[-1]][\mathcal{J}] = [\mathcal{J}].$$



- $P_{[x]}[-1][\mathcal{J}] = [\{x\} \cup \mathcal{J}]$, since we are looking at

$$\{(\mathcal{J}' \supseteq \mathcal{J}, x) : \text{supp}(\mathcal{J}'/\mathcal{J}) = \{x\}\} \subset X^{[n]} \times X^{[n+1]} \times X.$$

- ▶ $P_{[X]}[1]P_{[x]}[-1][\mathcal{J}]$, since

$$\{(\mathcal{J}'' \subseteq \mathcal{J}', x) : \text{supp}(\mathcal{J}'/\mathcal{J}'') = \{y\}, \text{ for some } y \in X\}$$

Easy case

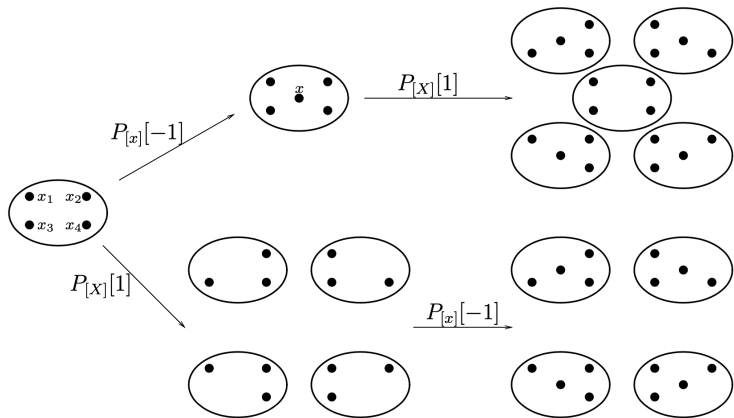
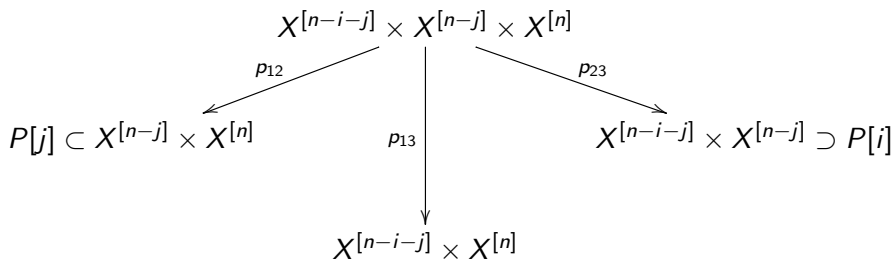


FIGURE 8.1. The proof of $[P_{[X]}[1], P_{[x]}[-1]][\mathcal{J}_1] = [\mathcal{J}_1]$

Proof for $i, j > 0$

Let's explain the case $i, j > 0$. By definition,

$$P_\alpha[i]P_\beta[j] = p_{13,*}(p_{12}^*[P[i]] \cap \Pi_1^*\alpha \cap p_{23}^*[P[j]] \cap \Pi_2^*\beta).$$



Proof for $i, j > 0$

$$\begin{aligned} P[i, j] &:= p_{12}^{-1}(P[i]) \cap p_{23}^{-1}(P[j]) \cap (X^{[n-i-j]} \times X^{[n-j]} \times X^{[n]}) \\ &= \left\{ (\mathcal{J}_1 \supseteq \mathcal{J}_2 \supseteq \mathcal{J}_3) : \begin{array}{l} \text{supp}(\mathcal{J}_1/\mathcal{J}_2) = \{x\} \text{ for some } x \in X \\ \text{supp}(\mathcal{J}_2/\mathcal{J}_3) = \{y\} \text{ for some } y \in X \end{array} \right\} \end{aligned}$$

Proof for $i, j > 0$

$$\begin{aligned} P[i, j] &:= p_{12}^{-1}(P[i]) \cap p_{23}^{-1}(P[j]) \cap (X^{[n-i-j]} \times X^{[n-j]} \times X^{[n]}) \\ &= \left\{ (\mathcal{J}_1 \supseteq \mathcal{J}_2 \supseteq \mathcal{J}_3) : \begin{array}{l} \text{supp}(\mathcal{J}_1/\mathcal{J}_2) = \{x\} \text{ for some } x \in X \\ \text{supp}(\mathcal{J}_2/\mathcal{J}_3) = \{y\} \text{ for some } y \in X \end{array} \right\} \end{aligned}$$

$P[i, j]$ is the disjoint union of $\{x \neq y\}$ and $\{x = y\}$.

Let $\bar{P}[i, j] := \overline{P[i, j] \cap \{x \neq y\}}$. Then

$$p_{12}^*[P[i]] \cap p_{23}^*[P[j]] = [\bar{P}[i, j]] + \iota_* R$$

where $R \in H_*^{BM}(P[i, j] \cap \{x = y\})$ corresponds to the $\{x = y\}$ part of $P[i, j]$ and ι is the inclusion.

Proof for $i, j > 0$

Lemma

1. *There exists an isomorphism*

$$P[i, j] \cap \{x \neq y\} \xrightarrow{\cong} P[j, i] \cap \{x \neq y\},$$

which exchanges Π_1 and Π_2 .

2. $p_{13,*}(\Pi_1^* \alpha \cap \Pi_2^* \beta \cap \iota_* R) = 0, \quad p_{13,*}(\Pi_1 \beta \cap \Pi_2^* \alpha \cap \iota_* R') = 0.$

Proof for $i, j > 0$

Lemma

1. *There exists an isomorphism*

$$P[i, j] \cap \{x \neq y\} \xrightarrow{\cong} P[j, i] \cap \{x \neq y\},$$

which exchanges Π_1 and Π_2 .

2. $p_{13,*}(\Pi_1^* \alpha \cap \Pi_2^* \beta \cap \iota_* R) = 0, \quad p_{13,*}(\Pi_1 \beta \cap \Pi_2^* \alpha \cap \iota_* R') = 0.$

This lemma implies the result since

$$\begin{aligned} P_\alpha[i] P_\beta[j] &= (-1)^{\deg \alpha} p_{13,*}((\overline{P}[i, j] + \iota_* R) \cap \Pi_1^* \alpha \cap \Pi_2^* \beta) \\ &= (-1)^{\deg \alpha \deg \beta} p_{13,*}((\overline{P}[j, i] + \iota_* R') \cap \Pi_1^* \beta \cap \Pi_2^* \alpha) \\ &= P_\beta[j] P_\alpha[i]. \end{aligned}$$

Proof of 1.

Define an isomorphism by

$$\mathcal{I}_1/\mathcal{I}_2' = \mathcal{I}_2/\mathcal{I}_3, \quad \mathcal{I}_2'/\mathcal{I}_3 = \mathcal{I}_1/\mathcal{I}_2.$$

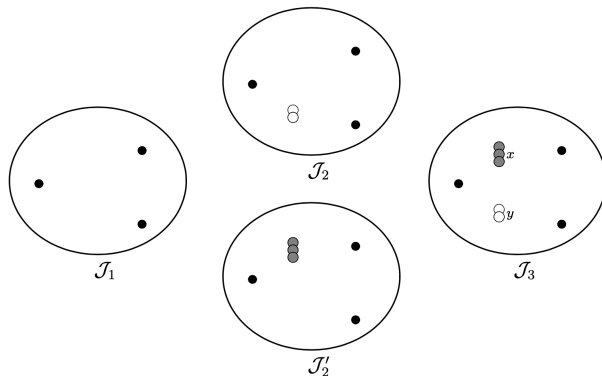


FIGURE 8.2. Correspondence between $\mathcal{I}_1 \supset \mathcal{I}_2 \supset \mathcal{I}_3$ and $\mathcal{I}_1 \supset \mathcal{I}_2' \supset \mathcal{I}_3$

Proof of 2.

Consider the commutative diagram

$$\begin{array}{ccc} P[i,j] \cap \{x = y\} & \xrightarrow{\Pi'} & X \\ \downarrow \iota & & \downarrow \Delta \\ P[i,j] & \xrightarrow{\Pi_1 \times \Pi_2} & X \times X. \end{array}$$

where Δ is the diagonal embedding and Π' is any of the projections Π_1 or Π_2 . Then

$$\begin{aligned} \Pi_1^* \alpha \cap \Pi_2^* \beta \cap \iota_* R &= \iota_* (\iota^* (\Pi_1^* \alpha \cap \Pi_2^* \beta) \cap R) \\ &= \iota_* (\iota^* (\Pi_1 \times \Pi_2)^* (\alpha \times \beta) \cap R) \\ &= \iota_* (\Pi'^* \Delta^* (\alpha \times \beta) \cap R) \\ &= \iota_* (\Pi'^* (\alpha \cap \beta) \cap R). \end{aligned}$$

Proof of 2.

Consider another commutative diagram:

$$\begin{array}{ccc}
 P[i, j] \cap \{x = y\} & \xrightarrow{\quad \Pi' \quad} & X \\
 \downarrow p_{13 \circ \iota} & & \parallel \\
 P[i + j] = \{(\mathcal{I}_1 \supseteq \mathcal{I}_3) : \text{supp}(\mathcal{I}_1 / \mathcal{I}_3) = \{x\}\} & \xrightarrow[\quad \Pi'' \quad]{} & X,
 \end{array}$$

where Π'' is the usual projection to X from the set $P[i + j]$. Then

$$p_{13, *}\iota_*(\Pi'^*(\alpha \cap \beta) \cap R) = \Pi''^*(\alpha \cap \beta) \cap p_{13, *}\iota_*(R).$$

Since $P[i + j]$ has less than the “expected dimension”,
 $p_{13, *}\iota_*(R) = 0$.

Bibliography

-  Nakajima, H. Lectures on Hilbert schemes of points on surfaces. In: Algebraic Geometry, Santa Cruz 1995, Proc. Sympos. Pure Math., vol. 62, AMS, 1997, pp. 33–104.
-  Nakajima, H. Heisenberg algebra and Hilbert schemes of points on projective surfaces. Ann. of Math. (2) 145 (1997), no. 2, 379–388.
-  Fogarty, J. Algebraic families on an algebraic surface. Amer. J. Math. 90 (1968), 511–521.
-  Fulton, W. Young Tableaux. Cambridge University Press, 1997.
-  Online videos: https://www.youtube.com/playlist?list=PLtmvIY4GrVv_tNl9B41l6k8Xl1k_RGwGO